Detecting Unknown Voice Forgeries via One-Class Learning and wav2vec 2.0

Yin and Lei (2026), in their paper *A Novel Attack-Resistant Speech Forgery Detection Method Integrating One-Class Learning and Wav2vec2*, present an approach for detecting voice forgeries that combines One-Class Learning with speech representations produced by a pre-trained Self-Supervised Learning model. The study addresses a central limitation of supervised Anti-Spoofing systems: dependence on labeled forgery examples, which may harm the ability to generalize to attacks based on synthesis and voice-conversion technologies not included in training. The researchers propose to cope with this problem by modeling the distribution of authentic speech and detecting recordings that deviate from it, instead of relying on learning features unique to known attacks.

The method is based on Deep Support Vector Data Description (Deep SVDD), a One-Class algorithm that maps real speech examples to a learned feature space and defines a decision boundary around them in the shape of a Hypersphere. During training the model seeks to create a compact representation of real speech, while at the evaluation stage the distance of a new recording from the center of the sphere is used for anomaly detection. Recordings located outside the decision boundary are classified as fake. The principal advantage of this approach is that one-class training does not require prior familiarity with every possible type of forgery, and therefore may be especially suited to detecting unknown attacks.

For feature extraction, the researchers combine two sources of information: classical acoustic features of the Linear Frequency Cepstral Coefficients (LFCC) type, and deep speech representations produced by means of wav2vec 2.0 Base. The representations produced by wav2vec 2.0 were learned through self-training on unlabeled speech data, and therefore may provide a rich description of spectral, temporal and contextual features that does not directly depend on forgery examples. In the configuration described, the Transformer layers of wav2vec 2.0 remain frozen, and an output layer is adapted. In parallel, the LFCC features include 64 basic coefficients alongside first- and second-order derivatives, for a total representation of 192 components. The sub-networks use Bi-LSTM and LCNN components and produce representations of dimension 256 for Deep SVDD.

In addition, the researchers propose a Multi-Model Weighted Fusion strategy that combines the scores of four sub-networks. The combination weights are determined by means of Optuna, while minimizing the Equal Error Rate (EER) metric on the Validation set. The combination is intended to exploit complementary features and reduce dependence on a single representation or a single sub-architecture. However, using EER for choosing the combination weights raises a methodological limitation: although Deep SVDD training is described as based on real speech alone, computing EER requires a validation set that also includes fake examples. Therefore the claim that the full system does not require labeled forgery examples is not fully supported by the protocol description.

The researchers evaluated the method by means of ASVspoof 2019 Logical Access, presented as a known-attacks scenario, and ASVspoof 2021 Deepfake, presented as an unknown-attacks scenario. In the first experiment the proposed method achieved an EER of 2.13%, compared with 2.35% for wav2vec 2.0 with a supervised binary classifier, 5.42% for Deep SVDD without wav2vec 2.0, and 8.76% for LFCC with One-Class SVM. In the second experiment, on ASVspoof 2021 DF, the method achieved an EER of 6.89%, compared with 18.45%, 15.67% and 22.31%, respectively. These findings indicate a significant advantage of the combined method in evaluating the unknown attacks within the reported experiments, especially compared with the binary classifier based on wav2vec 2.0.

However, the findings should be interpreted with caution. The comparison between ASVspoof 2019 LA and ASVspoof 2021 DF also includes changes in the data source, the attack types and the evaluation conditions, and therefore does not fully isolate the contribution of the novelty in the attack type. In addition, the method combines several components, including wav2vec 2.0, Deep SVDD and Weighted Fusion, so that the entire improvement cannot be attributed to One-Class Learning alone. The Ablation experiments are described as demonstrating the contribution of the different components and the possible improvement by means of Data Augmentation, but there are inconsistencies between the configuration names and the EER values in the table and the explanations in the text. Accordingly, not all the values can be relied upon for an exact quantification of the independent contribution of each component.

An additional limitation stems from the very nature of one-class learning. A real speech example recorded under conditions different from those included in training may fall outside the learned distribution region and be mistakenly classified as fake. Therefore detecting a deviation is not necessarily equivalent to detecting forgery. This problem is especially significant in evaluation between different data sources, where differences in speakers, language, microphone and recording conditions may affect the distances of the representations. In addition, the paper does not provide a sufficiently detailed description of all the stages of creating the validation sets and the final model selection, and does not present a direct evaluation on ASVspoof 5 or CodecFake.

For the present study, the central contribution of the paper is in presenting One-Class Learning as an alternative to binary classification based on labeled forgeries. Given the course dataset, which includes real and fake recordings from ASVspoof 5 and from CodecFake, alongside real speech examples from M-AILABS, one can examine using pre-trained wav2vec 2.0 or WavLM representations for modeling real speech. A suitable experiment would compare, under a shared evaluation protocol, a binary classifier using both classes against a One-Class approach based on real recordings alone at the stage of training the anomaly classifier. The evaluation will be based on EER and on error analysis by data source, while examining the risk of wrongly rejecting authentic speech coming from a source not included in training. It is recommended to start with a simple approach, such as a One-Class SVM on frozen wav2vec 2.0 representations, and only afterwards consider implementing Deep SVDD and combining several sub-networks, according to the results and the available resources.

Bibliographic source:

Yin, J., & Lei, S. (2026). *A Novel Attack-Resistant Speech Forgery Detection Method Integrating One-Class Learning and Wav2vec2*. In 2026 International Conference on Communication Networks and Machine Learning (CNML) (pp. 415–419). IEEE. https://doi.org/10.1109/CNML68938.2026.11452384